OPTICS · SEMESTER 5 · MODULE 1

Prepare: Fermat, curved surfaces & thin lenses

Everything you need before you open the question paper, from Plus Two up to Ghatak §3.1–3.2 and §4.1–4.7.

File 1 of 3	**Prepare** (this file): learn the ideas
File 2	Model paper: PYQs + new questions
File 3	Answer key: full solutions

Each answer in File 3 starts with a list of topic codes, for example T7 P4 M1. If a code is new to you, read that topic here first.

How to read this file

Read one topic at a time, in order. Each topic builds only on the ones before it.

REMIND ME

Facts from Class 10, Plus One or Plus Two that you have probably forgotten. Read these even if you think you know them.

MATHS TOOL

A small piece of maths, explained the first time it is needed.

THE IDEA

The physics in plain words, before any symbols.

RESULT

The formula or statement you must remember and write in the exam.

EXAM TRAP

A mistake that students make often, and how to avoid it.

CHECK YOURSELF

Questions to try. The answer is printed just below each one. Cover it with your thumb and try first.

The colour code

Every diagram in all three files uses the same colours, so you can read a picture quickly.

— blue	incoming (incident) ray
— orange	reflected ray
— green	refracted ray (after it enters the new material)
- - grey	normal, or the axis
dashed ray	a ray extended backwards. Light does not really travel along it.
▲ **black arrow**	object
▲ pink arrow	image (dashed if virtual)

Topic codes

P1–P4	Plus Two physics bridge
M1–M6	Maths tools
T1–T15	Module 1 topics
Q12	question 12 of the model paper (File 2)

The map of Module 1

SYLLABUS UNIT	TOPICS
1. Laws of reflection and refraction from Fermat's principle	T1–T5
2. Refraction and reflection at one spherical surface	T6–T10
3. Thin lens, principal foci, focal length	T11–T13
4. Newton formula, lateral magnification	T14–T15

The syllabus gives this module 15 of the 70 external marks. In the Nov 2024 and Nov 2025 papers it gave 2-mark definitions and one 5-mark numerical. The older 80-mark papers also asked 4-mark derivations.

The whole module in one story

THE IDEA

1. Light goes from A to B along a path where the travel time is "stationary": a tiny change of path does not change the time (T1–T2).

2. From that one rule we get three facts: light goes straight, angle in = angle out at a mirror, and Snell's law at a boundary (T3–T5).

3. Apply Snell's law at one curved glass surface. For rays close to the axis, all rays from a point O meet again at one point I. This gives one formula (T6–T9).

4. A mirror is the same calculation. Only the formula's details change (T10).

5. A lens is two curved surfaces. Use the formula twice to get the lens formula and the focal length (T11–T13).

6. Measure distances from the foci instead of the lens, and you get Newton's formula. Compare image size with object size to get magnification (T14–T15).

So the module is **one principle** (Fermat), **one approximation** (small angles) and **one sign rule**, used again and again.

P1 Light, rays and refractive index

What a "ray" is, and why light slows down in glass.

REMIND ME · CLASS 10 / PLUS TWO CH 9

- A **ray** is a line that shows which way light energy travels. A **beam** is a bundle of rays.
- Light travels at $c = 3 \times 10^8$ m/s in vacuum. Inside a material it is slower, with speed v.
- **Refractive index** $n = \dfrac{c}{v}$. Bigger n means slower light. Air: $n \approx 1$. Water: 1.33. Glass: about 1.5.
- A material with larger n is called **optically denser**.

THE IDEA

Think of n as a "slowness number". In glass ($n = 1.5$) light is 1.5 times slower than in vacuum, so 1 cm of glass takes as long to cross as 1.5 cm of vacuum.

1. Speed in glass: $v = \dfrac{c}{n} = \dfrac{3 \times 10^8}{1.5} = 2 \times 10^8$ m/s

   Rearranged $n = c/v$.

2. Time to cross 2 cm of glass: $t = \dfrac{0.02}{2 \times 10^8} = 10^{-10}$ s

   time = distance ÷ speed.

3. Vacuum distance in the same time: $ct = 3 \times 10^8 \times 10^{-10} = 0.03$ m $= 3$ cm $= n \times 2$ cm
   So "n × length" measures time, written as a distance. This is the **optical path** of T1.

CHECK YOURSELF

1. Speed of light in water ($n = 1.33$)?

$v = 3 \times 10^8 / 1.33 \approx 2.26 \times 10^8$ m/s.

2. Which is slower for light: a medium with $n = 1.6$ or $n = 1.4$?

$n = 1.6$. Bigger n means slower light.

P2

Laws of reflection and refraction

The two laws that Fermat's principle will later

explain.

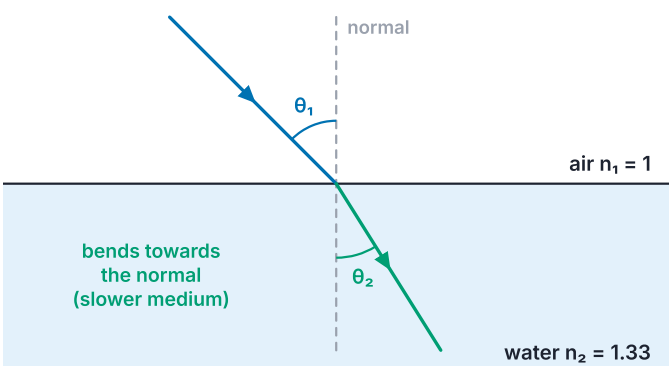
REMIND ME · CLASS 10

Reflection: (1) the incident ray, the reflected ray and the normal lie in one plane; (2) angle of incidence = angle of reflection, $i = r$.

Refraction (Snell's law): (1) the incident ray, the refracted ray and the normal lie in one plane; (2) $n_1 \sin\theta_1 = n_2 \sin\theta_2$.

The **normal** is the line at 90° to the surface at the point where the ray hits it.

In Class 10 you wrote Snell's law as $\frac{\sin i}{\sin r} = n_{21} = \frac{n_2}{n_1}$. Cross-multiply and you get $n_1 \sin i = n_2 \sin r$: the same law.



Refraction. Angles are measured from the **normal** (grey dashed), never from the surface. $n_1 \sin\theta_1 = n_2 \sin\theta_2$.

THE IDEA

Going into a slower material (bigger n), light bends **towards** the normal. Coming out into a faster one, it bends **away**. A ray along the normal ($\theta_1 = 0$) does not bend at all.

EXAM TRAP

All angles are measured from the **normal**, never from the surface. If a question gives the angle with the surface, subtract it from 90° first.

CHECK YOURSELF

1. Light goes from air into glass ($n = 1.5$) at 30° to the normal. Find the refracted angle.

$\sin\theta_2 = \dfrac{1 \times \sin 30^\circ}{1.5} = \dfrac{0.5}{1.5} = 0.333$, so $\theta_2 \approx 19.5^\circ$. It bends towards the normal, as expected.

2. A ray makes 60° with a mirror's surface. What is the angle of incidence?

$90^\circ - 60^\circ = 30^\circ$.

P3 Mirrors and lenses from Plus Two

The words and formulas you met in Plus Two, all in

one place.

REMIND ME · WORDS

Pole P	centre of the mirror or surface (for a lens: the optical centre)
Centre C	centre of the sphere the surface is cut from
Radius R	distance PC
Principal axis	the line through P and C
Focus F	where rays parallel to the axis meet (or seem to come from) after the mirror or lens
Focal length f	distance PF. For a mirror, $f = R/2$.

REMIND ME · REAL VS VIRTUAL IMAGE

Real image: the rays really meet there. You can catch it on a screen. Usually upside down.

Virtual image: the rays only *seem* to come from there when you trace them backwards (dashed lines). No screen can catch it, but your eye sees it. Example: your face in a plane mirror.

REMIND ME · PLUS TWO FORMULAS (NCERT)

Mirror	$\frac{1}{v}+\frac{1}{u}=\frac{1}{f}, \quad m=-\frac{v}{u}$
Lens	$\frac{1}{v}-\frac{1}{u}=\frac{1}{f}, \quad m=\frac{v}{u}$
Lens maker	$\frac{1}{f}=(n-1)\left(\frac{1}{R_1}-\frac{1}{R_2}\right)$
Power	$P=\frac{1}{f}$ (f in metres), unit dioptre (D)
Curved surface	$\frac{n_2}{v}-\frac{n_1}{u}=\frac{n_2-n_1}{R}$

Good news: **Ghatak uses the same sign convention as NCERT**, so all of these stay true. In this module you learn *where they come from*.

Convex lens: where is the image? (Plus Two table)

OBJECT AT	IMAGE AT	IMAGE IS
very far (∞)	at F_2	real, tiny point
beyond $2F_1$	between F_2 and $2F_2$	real, inverted, smaller
at $2F_1$	at $2F_2$	real, inverted, same size
between F_1 and $2F_1$	beyond $2F_2$	real, inverted, bigger
at F_1	at ∞	(rays come out parallel)
between F_1 and lens	same side as object	virtual, upright, bigger (magnifying glass)

A concave mirror follows the same pattern, but its image forms in front of the mirror (on the object's side).

CHECK YOURSELF

1. You hold a magnifying glass close to a letter. Is the image real or virtual?

Virtual and upright. The object is between the lens and F_1.

2. Concave mirror, $R = 30$ cm. Size of f?

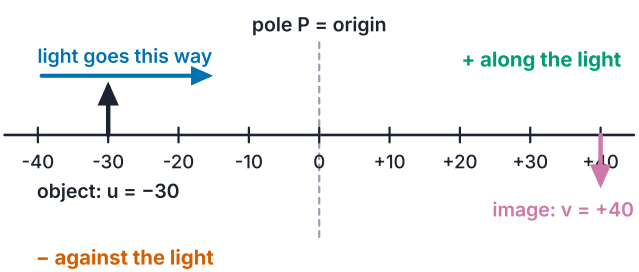
$|f| = R/2 = 15$ cm.

P4 The sign rule (Cartesian convention)

> The source of most lost marks. Learn it once, properly.

THE IDEA

Treat the axis like a number line. Put **zero at the pole** (or at the lens). Draw light going **left to right**. Then each distance is a *coordinate*: its sign tells you which side it is on.



Cartesian sign rule. Every distance is measured **from the pole**. Same direction as the light → plus. Opposite → minus. Heights: up plus, down minus.

RESULT · THE FOUR RULES (GHATAK §4.2.1 = NCERT)

1. Measure all distances **from the pole** P (or the centre of the lens).
2. Distances in the direction the light travels (to the right) are **+**. Against it (to the left) are **−**.
3. Heights above the axis are **+**. Below are **−**.
4. R is **+** if the centre C is to the right of P, and **−** if C is to the left.

So a real object (on the left, where light starts) always has u **negative**.

The sign rule in four quick pictures

Object 30 cm left of lens	$u = -30$ cm
Image 60 cm right of lens	$v = +60$ cm (real, for a lens)
Image 20 cm left of lens	$v = -20$ cm (virtual, for a lens)
Glass bulging towards the light	C is on the right, so $R > 0$
Concave mirror facing the light	its hollow faces the light, and the centre of a hollow is in front of it, on the light's side. So C is on the left: $R < 0$ and $f = R/2 < 0$
Image below the axis	height $h' < 0$ (inverted)

EXAM TRAP

Write each value **with its sign** before you put it in a formula. Write $u = -30$, not "u = 30". The formula already knows the directions. Your job is only to give it the coordinates.

Some books use a "real is positive" rule. Its formulas look different, for example

$$\frac{n_1}{u} + \frac{n_2}{v} = \frac{n_2 - n_1}{R}.$$

Do not mix the two. Ghatak and this file use only the Cartesian rule.

CHECK YOURSELF

1. Lens formula with $u = -30$, $f = +20$. Find v.

$\frac{1}{v} = \frac{1}{f} + \frac{1}{u} = \frac{1}{20} - \frac{1}{30} = \frac{3-2}{60} = \frac{1}{60}$, so $v = +60$ cm: real, on the right.

2. A biconvex lens. Signs of R_1 (first surface) and R_2 (second surface)?

First surface bulges left, so its centre is on the right: $R_1 > 0$. Second surface bulges right, so its centre is on the left: $R_2 < 0$.

M1 Radians and the small-angle trick

Used in every curved-surface derivation. It turns

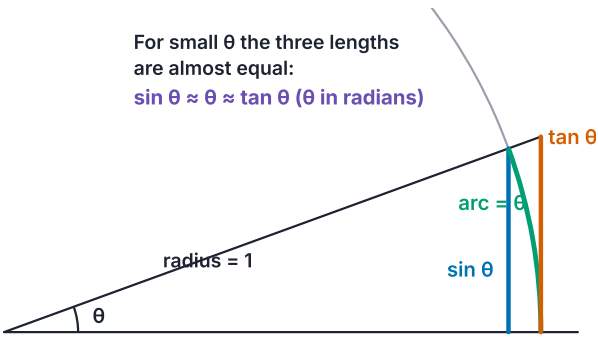
hard trigonometry into simple fractions.

REMIND ME · PLUS ONE

An angle in **radians** = arc length ÷ radius. A full turn is 2π rad = 360°, so $1^\circ = \pi/180 \approx 0.0175$ rad.

In a right triangle:

$$\sin\theta = \frac{\text{opposite}}{\text{hypotenuse}}, \qquad \tan\theta = \frac{\text{opposite}}{\text{adjacent}}$$



Why the small-angle trick works: blue, green and orange are nearly the same length when θ is small. At θ = 0.1 rad: sin = 0.0998, tan = 0.1003.

RESULT · SMALL-ANGLE APPROXIMATION

If θ is small (below about 10°, i.e. 0.17 rad) **and in radians**:

$$\sin\theta \approx \theta, \qquad \tan\theta \approx \theta, \qquad \cos\theta \approx 1 - \frac{\theta^2}{2}$$

HOW WE USE IT

A ray hits a surface at height h, at distance x along the axis. Its angle with the axis is α, with $\tan\alpha = h/x$. For small angles we just write

$$\alpha \approx \frac{h}{x}.$$

Nearly every angle in T7, T10 and T15 is written this way.

EXAM TRAP

Never use degrees: $\sin 5^\circ \neq 5$. And the trick works only for rays close to the axis. That condition is called the **paraxial approximation** (T6).

CHECK YOURSELF

1. A ray hits a surface 2 cm above the axis, at a point 40 cm from the object. Its angle with the axis?

$\alpha \approx 2/40 = 0.05$ rad (about 2.9°).

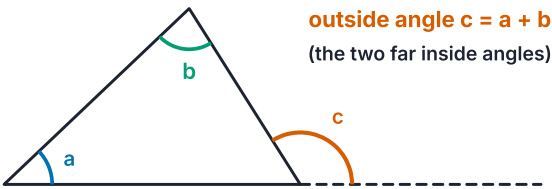
M2 Geometry tools

> Four facts from Class 9–10 that every derivation uses.

1. Pythagoras

A right triangle with sides a, b has hypotenuse $\sqrt{a^2+b^2}$. Light going from height h down to a point at horizontal distance x travels $\sqrt{x^2+h^2}$.

2. Exterior angle rule



Exterior angle rule (Class 9). We use it on every curved-surface diagram.

Why: $a + b + (\text{third angle}) = 180^\circ$, and $c + (\text{third angle}) = 180^\circ$ because they form a straight line. So $c = a + b$.

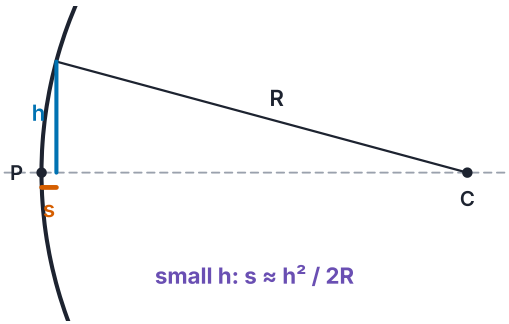
3. Similar triangles

Two triangles with the same angles have the same shape, so their sides are in the same ratio. If a small triangle has height h and base u, and a similar big one has height h' and base v:

$$\frac{h'}{h} = \frac{v}{u}$$

This gives every magnification formula (T14, T15).

4. Sagitta: how much a curve bulges



Sagitta: how far a curved surface bulges (s) at height h. From Pythagoras $R^2 = h^2 + (R - s)^2$ and dropping the tiny s^2.

1. $R^2 = h^2 + (R - s)^2$
   Pythagoras in the right triangle with sides h, $R - s$ and hypotenuse R.

2. $R^2 = h^2 + R^2 - 2Rs + s^2$
   Expand the bracket.

3. $2Rs \approx h^2$, so $s \approx \dfrac{h^2}{2R}$
   Cancel R^2. s is tiny, so s^2 is even tinier: drop it.

CHECK YOURSELF

1. Triangle angles 50° and 60°. The exterior angle next to the third angle?

$50^\circ + 60^\circ = 110^\circ$.

2. A lens surface with $R = 50$ cm, measured 1 cm from the axis. How much does it bulge?

$s \approx 1^2/(2 \times 50) = 0.01$ cm. Very small, which is why we can treat the surface as flat when we measure distances.

M3 Slope zero: finding a stationary point

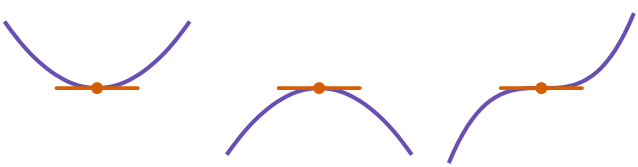
> The calculus inside Fermat's principle. You need only one idea: at the bottom of a valley the ground is flat.

REMIND ME · PLUS ONE / PLUS TWO MATHS

The derivative $\frac{dL}{dx}$ is the **slope** of the graph of L against x: how fast L changes when x changes a little.

Chain rule: $\frac{d}{dx}\sqrt{g(x)} = \frac{g'(x)}{2\sqrt{g(x)}}$.

orange = flat spot: slope = 0 = "stationary"



Three kinds of stationary point. Fermat says the light path is at one of these. Usually it is the valley (least time), but not always.

DERIVATIVES FROM ZERO (IF YOU NEVER LEARNT THEM)

Slope = (change in L) ÷ (change in x), for a **very tiny** change. Try $L = x^2$ at $x = 3$: $L(3.01) - L(3) = 9.0601 - 9 = 0.0601$. Divide by the change 0.01: slope $\approx 6.01 \approx 6$.

The rule $\dfrac{d}{dx}x^n = n\,x^{n-1}$ gives $2x = 6$ at once. A constant has slope 0. A sum: add the slopes.

Chain rule in words: slope of the outside × slope of the inside. For $\sqrt{g}$: outside slope $\dfrac{1}{2\sqrt{g}}$, inside slope g'.

THE IDEA

Where the graph is flat, a tiny change in x makes **almost no change** in L. This is a **stationary point**. To find it, set $\dfrac{dL}{dx} = 0$. It can be a minimum, a maximum or a flat shelf. The second derivative tells which: $L'' > 0$ means a valley (minimum), $L'' < 0$ a hilltop (maximum).

The one derivative you must know

1. $$\frac{d}{dx}\sqrt{x^2 + a^2} = \frac{2x}{2\sqrt{x^2 + a^2}} = \frac{x}{\sqrt{x^2 + a^2}}$$
   Chain rule with $g = x^2 + a^2$, $g' = 2x$. The 2s cancel.

2. $$\frac{d}{dx}\sqrt{(d - x)^2 + b^2} = \frac{-(d - x)}{\sqrt{(d - x)^2 + b^2}}$$
   Now $g' = 2(d - x) \times (-1)$. Note the **minus**: as x grows, $d - x$ shrinks.

READ IT AS A SINE

In a right triangle with horizontal side x and vertical side a, the hypotenuse is $\sqrt{x^2+a^2}$. So $\dfrac{x}{\sqrt{x^2+a^2}}$ is opposite/hypotenuse **for the angle measured from the vertical**, which is the normal. That is $\sin\theta$. This is how Snell's law appears in T3.

CHECK YOURSELF

1. Where is $L(x)=(x-3)^2+5$ stationary? Minimum or maximum?

$L'=2(x-3)=0$ gives $x=3$. $L''=2>0$, so it is a minimum (valley), with $L=5$.

2. Derivative of $\sqrt{x^2+16}$ at $x=3$?

$3/\sqrt{25}=0.6$.

M4 The binomial shortcut

Square roots of "1 + something tiny". Used in the Fermat route for curved surfaces (T8, T10).

RESULT

For small ε:

$$(1+\varepsilon)^n \approx 1+n\varepsilon, \qquad \text{so} \qquad \sqrt{1+\varepsilon}\approx 1+\frac{\varepsilon}{2}$$

Why it works: $(1+\varepsilon)^2=1+2\varepsilon+\varepsilon^2\approx 1+2\varepsilon$, because ε^2 is tiny. Going backwards, $\sqrt{1+2\varepsilon}\approx 1+\varepsilon$, which is the same as $\sqrt{1+\varepsilon}\approx 1+\dfrac{\varepsilon}{2}$.

1. Check with numbers: $\sqrt{1.02} = 1.00995\ldots$ and $1 + 0.02/2 = 1.01$

   Almost the same. The error is about $\varepsilon^2/8$.

2. Typical use: $\sqrt{x^2 + k\theta^2} = x\sqrt{1 + \dfrac{k\theta^2}{x^2}} \approx x + \dfrac{k\theta^2}{2x}$

   Take the big part x^2 outside the root. What is left is $1+$ (tiny), then use the shortcut.

REMIND ME

Keep terms up to θ^2 and drop θ^4 and higher. That is the whole "paraxial" game.

M5 Ellipse and parabola: the focus property

> Two curves whose shape makes all light paths equal. Used in T5.

REMIND ME · PLUS ONE CONIC SECTIONS

Ellipse: the set of points P whose distances to two fixed points F_1, F_2 (the foci) add up to a constant: $PF_1 + PF_2 = 2a$. Draw it with two pins and a loop of string.

Parabola: the set of points equally far from one point (the focus S) and one line (the directrix).

NUMBERS MAKE IT REAL

Take an ellipse with $2a = 10$ and foci 8 apart. At the end of the long axis: distances 1 and 9, sum 10. At the top: distances 5 and 5, sum 10. The split changes but **the total stays the same**.

EXAM TRAP

It is the **sum** of the two distances that is constant. Each distance on its own changes from point to point.

M6 Working with reciprocals

How to solve $\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$ without mistakes.

1. Make the unknown alone: $\frac{1}{v} = \frac{1}{f} + \frac{1}{u}$
   Move $-\frac{1}{u}$ to the other side. It becomes $+\frac{1}{u}$.

2. Put numbers **with signs**: $f = +20$, $u = -30$ gives $\frac{1}{v} = \frac{1}{20} + \frac{1}{-30} = \frac{1}{20} - \frac{1}{30}$
   The sign of u goes inside the fraction.

3. Common denominator: $\frac{3}{60} - \frac{2}{60} = \frac{1}{60}$
   LCM of 20 and 30 is 60.

4. Flip only at the end: $v = +60$ cm
   Combine everything into one fraction first, then flip.

EXAM TRAP

$\frac{1}{v} = \frac{1}{20} - \frac{1}{30}$ does **not** give $v = 20 - 30$. Never flip the fractions one by one.

REMIND ME

On a calculator use decimals: $1/20 - 1/30 = 0.05 - 0.0333 = 0.01667$, and $1/0.01667 = 60$.

CHECK YOURSELF

Solve $\frac{1}{v} = \frac{1}{-90} + \frac{1}{50}$.

$\frac{-5+9}{450} = \frac{4}{450}$, so $v = 112.5$.

UNIT 1 · GHATAK §3.1–3.2

Fermat's principle

One rule about *time* explains straight lines, mirrors and refraction.

T1 Rays and optical path

> When can we draw light as lines? And how do we measure a path in "time units"?

THE IDEA · WHAT IS A RAY?

Shine a lamp through a 1 cm hole: you get a sharp patch of light. Make the hole tiny (about 0.1 mm) and the patch spreads out with fuzzy edges. That spreading is **diffraction**, and it happens because light is a wave.

When everything (holes, lenses, mirrors) is much bigger than the wavelength (about 5×10^{-5} cm), the spreading is too small to notice. Then light goes in sharp lines called **rays**. Studying light with rays is **geometrical optics**. It is the limit $\lambda \to 0$.

Ghatak §3.1. In the model paper: Q5.

Optical path = refractive index × length

1. Speed in a medium: $v = \dfrac{c}{n}$

   From P1.

2. Time for a small piece of path ds: $d\tau = \dfrac{ds}{v} = \dfrac{n\,ds}{c}$

   time = distance ÷ speed.

3. Add up all pieces from A to B: $\tau = \dfrac{1}{c}\displaystyle\int_A^B n\,ds$

   The sign ∫ is a stretched S, for "Sum". The integral just adds up all the tiny pieces (Ghatak eq. 3.1).

4 Name the integral: $L_{op} = \int_A^B n\,ds = c\tau$

So optical path = c × travel time. It is the distance light would cover **in vacuum** in the same time.

RESULT · OPTICAL PATH

Optical path length = (refractive index) × (geometrical path length). For several media: $L_{op} = n_1 l_1 + n_2 l_2 + \cdots$

Comparing optical paths is the same as comparing travel times.

EXAM TRAP

A shorter route through glass can have a **longer** optical path than a longer route through air. Always multiply by n before you compare.

CHECK YOURSELF

Light crosses 3 cm of air and then 2 cm of glass ($n = 1.5$). Optical path?

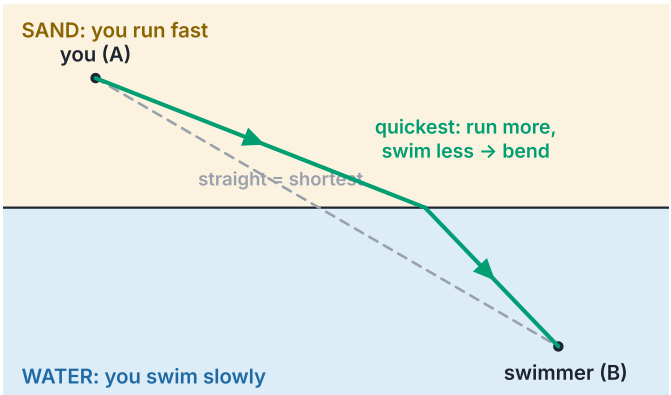
$1 \times 3 + 1.5 \times 2 = 6$ cm. Light takes as long as it would to cross 6 cm of vacuum.

In the model paper: Q1 (PYQ N20 Q1).

T2 Fermat's principle: stationary, not "least"

> The one rule behind the whole module, stated the

way examiners want.



The lifeguard problem. The quickest route bends at the shore. Light does the same at a surface. This bending *is* refraction.

THE IDEA

A lifeguard runs fast on sand and swims slowly in water. The quickest route to the swimmer is **not** the straight line. It is a bent line: run a bit further on sand, swim a bit less. Light behaves like the lifeguard. Each medium has its own speed, and the light's path bends where the medium changes.

RESULT · FERMAT'S PRINCIPLE (EXAM STATEMENT)

The actual path of a ray between two points is the path for which the optical path length (equivalently, the time taken) is **stationary**: it is a minimum, a maximum, or stays unchanged for small changes of the path.

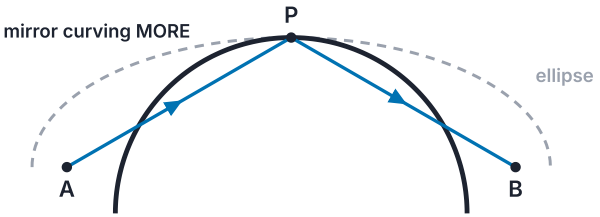
$$\delta \int_A^B n\, ds = 0 \qquad \text{(Ghatak 3.3)}$$

Read δ as "the change when the path is wiggled a little". The equation says: wiggle the path, and the optical path does not change (to first order).

"Stationary" means: move the path a tiny bit, and the optical path changes only by a **second-order** amount. That is the "flat spot" of M3.

Why not just "least time"?

Fermat first stated it as the principle of *least* time. Ghatak calls that form "incomplete and slightly incorrect", because some real rays take a path of **maximum** time, and some take a path whose time does not change at all.



path A→P→B is LONGER than nearby paths = maximum

A mirror that curves more sharply than the ellipse touching it at P: the real ray APB is a **maximum**. That is why Fermat says "stationary", not "least".

Minimum	straight line in one medium; plane-mirror reflection; ordinary refraction
Constant	elliptical mirror, focus to focus (T5)
Maximum	a mirror curving more sharply than the ellipse touching it (above); also points beyond the image of a curved surface (T8)

EXAM TRAP

Writing "light takes the shortest path" loses marks twice. The rule is about optical path (time), not length, and it is **stationary**, not least. Say "extremum or stationary" (PYQ N22 Q11 uses the word "extremum").

CHECK YOURSELF

1. State Fermat's principle in one line.

A ray travels between two points along the path whose optical path length ∫n ds is stationary (minimum, maximum or constant) compared with nearby paths.

2. Give one case where the time is a maximum.

Reflection inside a concave mirror that curves more sharply than the ellipse (with foci at the two points) touching it at the point of reflection.

In the model paper: **Q2** (PYQ N22 Q11), **Q39** .

T3 Result 1 and 2: straight lines and the law of reflection

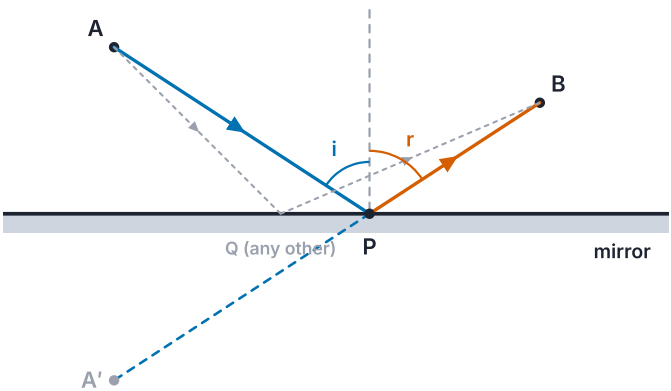
> The first two of the "three results obtained from Fermat's principle" (PYQ N20 Q11).

Result 1: in one medium, light goes straight

1. n is the same everywhere, so $L_{op} = \int n\, ds = n \times (\text{length})$
   n comes out of the integral because it is constant.
2. Stationary optical path = stationary length
   Multiplying by a constant n does not move the flat spot.

3. The shortest line between two points is straight, so the ray is a straight line (a minimum)
   This is **rectilinear propagation**.

Result 2: law of reflection



Fermat → law of reflection. A′ is the mirror image of A, so AP = A′P. The path A′PB is shortest when it is a **straight line**, which forces $i = r$. Any other point Q gives a longer path.

Light goes from A to B by touching the mirror at some point P. Which P?

1. Draw A′, the mirror image of A: as far below the mirror as A is above it
   A trick to "unfold" the bent path.

2. For every point P on the mirror, $AP = A'P$
   The mirror is the perpendicular bisector of AA′, so every point on it is equally far from A and A′.

3. So path $AP + PB = A'P + PB$
   Same length, now written as a path from A′ to B.

4. Shortest $A'P + PB$: when A′, P, B lie on **one straight line**

   Result 1. Any other point Q gives a longer path (grey dotted).

5. That straight line lies in the plane containing A, A′ and B, which also contains the normal at P

   AA′ is parallel to the normal. So incident ray, normal and reflected ray are in **one plane**.

6. (angle of PB with the upward normal) = (angle of PA′ with the downward normal)

   A′PB is one straight line crossing the normal line at P, so these are **vertically opposite angles**. The left side is r.

7. (angle of PA′ with the downward normal) = (angle of PA with the upward normal) $= i$

   A′ is the mirror copy of A, so the picture below the mirror is a flipped copy of the picture above.

8. Therefore $i = r$

   Both are equal to the same angle.

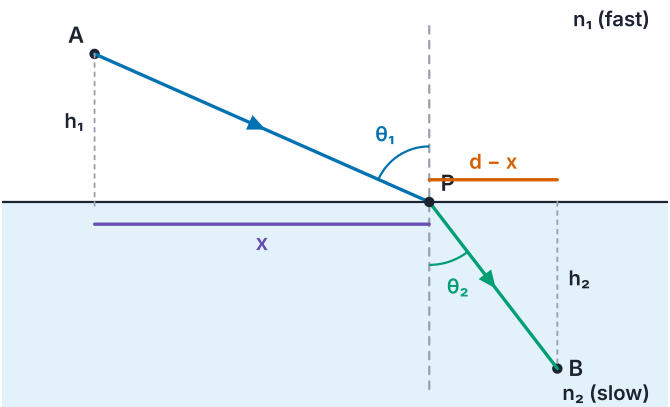
RESULT · LAW OF REFLECTION

The incident ray, the reflected ray and the normal lie in one plane, and $i = r$. Here the path is a **minimum**.

In the model paper: **Q3** (PYQ N20 Q11), **Q28** .

T4 Result 3: Snell's law from Fermat

> The lifeguard problem done with maths. A favourite derivation.



Fermat → Snell. Slide P along the surface (change x). The optical path $n_1 \cdot AP + n_2 \cdot PB$ is stationary when $n_1 \sin\theta_1 = n_2 \sin\theta_2$.

A is at height h_1 above the surface in medium n_1. B is at depth h_2 in medium n_2. Their horizontal separation is d (fixed). The ray crosses the surface at P, a horizontal distance x from A. Only x can change.

1. $AP = \sqrt{x^2 + h_1^2}, \quad PB = \sqrt{(d-x)^2 + h_2^2}$
   Pythagoras (M2).

2. $L(x) = n_1\sqrt{x^2 + h_1^2} + n_2\sqrt{(d-x)^2 + h_2^2}$
   Optical path = n × length for each part (T1). Ghatak eq. 3.4.

3. $$\frac{dL}{dx} = \frac{n_1 x}{\sqrt{x^2 + h_1^2}} - \frac{n_2(d-x)}{\sqrt{(d-x)^2 + h_2^2}} = 0$$
Fermat: slope zero (M3). Note the minus from the chain rule.

4. $$\frac{x}{\sqrt{x^2 + h_1^2}} = \sin\theta_1, \quad \frac{d-x}{\sqrt{(d-x)^2 + h_2^2}} = \sin\theta_2$$
Read off the two triangles: opposite ÷ hypotenuse, with the angles measured from the normal.

5. $n_1 \sin\theta_1 = n_2 \sin\theta_2$
Put these into step 3. This is **Snell's law** (Ghatak 3.6).

ALSO NOTE

- The coplanarity part follows as for reflection. If P moved out of the plane containing A, B and the normal, both AP and PB would get longer.
- The second derivative is positive, so this ray takes the **minimum** time.
- Fermat (1657) predicted that light bends **towards** the normal because it is **slower** in the denser medium. The particle model of that time predicted the opposite.

RESULT · THE THREE RESULTS FROM FERMAT (PYQ N20 Q11)

1. In a homogeneous medium light travels in straight lines (rectilinear propagation).
2. Law of reflection: coplanar, $i = r$.
3. Law of refraction: coplanar, $n_1 \sin\theta_1 = n_2 \sin\theta_2$.

EXAM TRAP

In step 3, students drop the minus sign. Then they get $n_1 \sin\theta_1 = -n_2 \sin\theta_2$ and get stuck. Check: when x grows, PB gets shorter, so its derivative must be negative.

CHECK YOURSELF

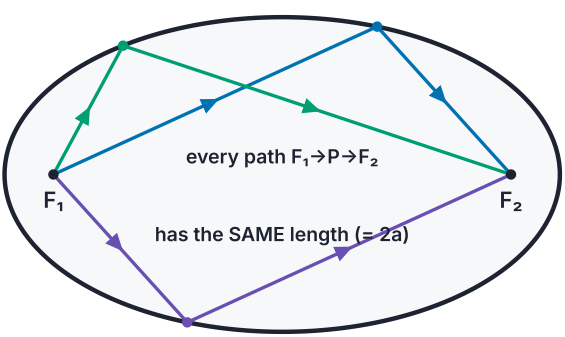
If both points are in the same medium ($n_1 = n_2$) and P is on a mirror instead, what does the same calculation give?

$\sin\theta_1 = \sin\theta_2$, so $i = r$: the law of reflection again.

In the model paper: Q29 , Q11 .

T5 Special mirrors and perfect images

> Ellipse, parabola, and why "equal optical paths" means a sharp image.

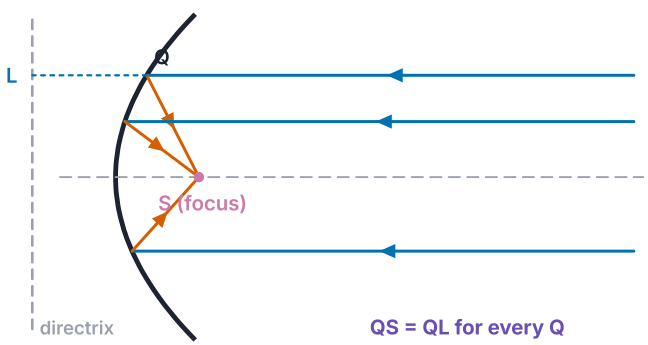


Elliptical reflector. Light from one focus, after reflecting anywhere, reaches the other focus. All the paths are equal, so the path is stationary (neither longer nor shorter than its neighbours).

ELLIPTICAL REFLECTOR (GHATAK EX. 3.2, PYQ N19 Q11)

Put a source at focus F_1 of a mirror shaped like an ellipse. For every point P on the mirror, $F_1P + PF_2 = 2a$ (M5). So every path from F_1 via the mirror to F_2 has **the same** optical path.

Then nearby paths are not longer and not shorter. The path is **stationary**: neither a minimum nor a maximum, but constant. Every ray from F_1 reaches F_2. This shows clearly why Fermat says "stationary" and not "least".



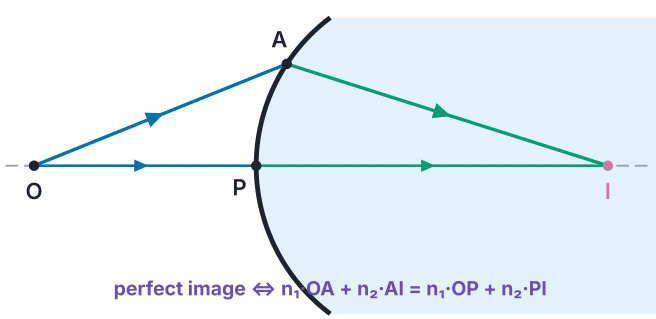
Paraboloidal mirror. Each point Q is as far from the focus S as from the directrix line. So every ray parallel to the axis travels the same optical path to S. All of them meet at S.

PARABOLOIDAL MIRROR (GHATAK EX. 3.1)

For rays coming in parallel to the axis, start every path at the same line P, perpendicular to the axis. Path = $PQ + QS = PQ + QL$, because a parabola has $QS = QL$ (M5). Now PQ and QL lie on one horizontal line, so $PQ + QL$ is just the distance from line P to the directrix. That is the same for every ray. So all parallel rays meet at the focus S, perfectly. This is why car headlights and dish antennas are paraboloids.

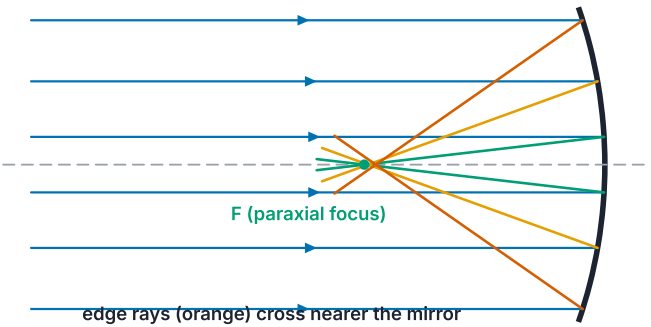
RESULT · PERFECT IMAGING

If **every** ray from a point O reaches a point I with the **same optical path**, they all arrive in step and I is a perfect (sharp) image of O. Fermat's principle is then satisfied for every ray at once.



Equal optical paths. If every ray from O reaches I with the same optical path, all of them arrive together and I is a sharp image.

Why a sphere is not perfect: spherical aberration



Spherical aberration. A spherical mirror sends rays at different heights to different points. Only the rays near the axis (paraxial, green) meet at the focus F.

A spherical mirror or surface makes the optical paths equal **only for rays close to the axis**. In T8 we will see the equality holds only up to terms in θ^2. Rays far from the axis cross it at slightly different points, so the image is blurred. This defect is **spherical aberration**. That is why every formula in this module needs the paraxial approximation (T6). A paraboloid has no such defect for parallel light.

CHECK YOURSELF

1. In an elliptical reflector, is the path from focus to focus a minimum?

No. All such paths are equal, so the path is stationary (constant), neither a minimum nor a maximum.

2. Why do big telescopes use paraboloidal mirrors instead of spherical ones?

A paraboloid brings all parallel rays to one focus: equal optical paths and no spherical aberration.

In the model paper: **Q4** (PYQ N19 Q11), **Q7**, **Q30**, **Q6**.

UNIT 2 · GHATAK §4.1–4.3 (+ §3.2 examples)

One curved surface

Snell's law at a curved piece of glass. Lenses and mirrors are built from this.

T6 Paraxial rays, axis and signs

The ground rules before any curved-surface formula.

RESULT · PARAXIAL APPROXIMATION (PYQ N22 Q1)

We use only rays that stay **close to the axis** and make **small angles** with it. These are **paraxial rays**. For them:

- $\sin\phi \approx \tan\phi \approx \phi$ (M1);
- the bulge of the surface is negligible (M2), so distances to the point where the ray hits can be measured from the pole P;
- all paraxial rays from one object point meet at one image point.

AXIS AND MERIDIONAL RAYS (GHATAK §4.2.2)

For a single spherical surface, the **axis** is the line joining the object point O and the centre C. Every ray from O lies in a plane that contains this line (a **meridional** plane), so after refraction it must cross the axis again. A ray that does not lie in such a plane is a **skew ray**.

For a lens, the axis is the line joining the centres of its two surfaces.

REMIND ME · THE SIGN RULE (P4)

Origin at the pole P. Light goes left → right. Right is +, left is −, up is +. $R > 0$ if C is to the right of P.

HOW GHATAK WORKS

In the derivations, Ghatak first uses plain **sizes** x, y, r (all positive) from the picture. At the end he replaces them with signed values: for a real object on the left, $u = -x$. This two-stage method is safe: geometry first, signs last. Use it in your answers too.

EXAM TRAP

The approximation fails for rays far from the axis. Their images are blurred: spherical aberration (T5). Say "paraxial" in every derivation of T7–T15. It earns a mark.

CHECK YOURSELF

1. Fill in: in the paraxial approximation we consider rays which pass _____ the axis.

close to (and at small angles with) the axis.

2. What is a meridional ray?

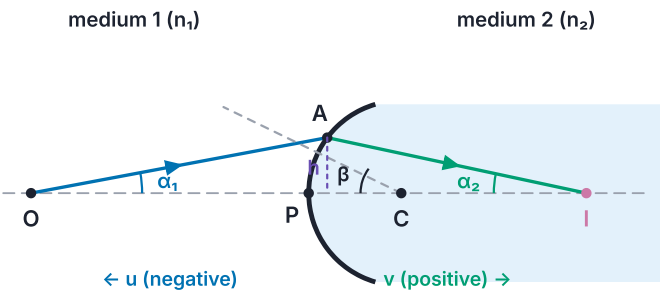
A ray lying in a plane that contains the axis. All rays from an axial object point are meridional for a single spherical surface.

In the model paper: **Q8** (PYQ N22 Q1), **Q9**, **Q10**.

T7 Refraction at one spherical surface (geometry route)

The most important derivation of the module (PYQ

> N19 Q18). Everything after this reuses it.



Refraction at one curved surface. O = object, P = pole, C = centre of curvature, I = image, A = where the ray hits, at height h. Grey dashed = normal (always through C).

O is a point object in medium n_1, at distance x from the pole P. The surface has centre C at distance r. A ray from O hits the surface at A, height h, and after refraction meets the axis at I, distance y from P. The normal at A is the line CA.

Angles: α_1 at O, β at C, α_2 at I (all with the axis). At A: angle of incidence ϕ_1, angle of refraction ϕ_2, both measured from the normal CA.

Step A: angles from geometry

1. $\phi_1 = \alpha_1 + \beta$
   Triangle OAC. ϕ_1 is the exterior angle at A, which equals the sum of the far angles at O and C (M2).

2. $\beta = \phi_2 + \alpha_2 \;\Rightarrow\; \phi_2 = \beta - \alpha_2$
   Triangle CAI. β is the exterior angle at C.

Step B: small angles

1. $\alpha_1 \approx \dfrac{h}{x}, \quad \beta \approx \dfrac{h}{r}, \quad \alpha_2 \approx \dfrac{h}{y}$

   Paraxial (T6): angle ≈ height ÷ distance along the axis (M1).

2. $\phi_1 \approx \dfrac{h}{x} + \dfrac{h}{r}, \quad \phi_2 \approx \dfrac{h}{r} - \dfrac{h}{y}$

   Put these into step A. These are Ghatak eqs. 4.2 and 4.3.

Step C: Snell's law, then cancel h

1. $n_1\phi_1 = n_2\phi_2$

   Snell $n_1 \sin\phi_1 = n_2 \sin\phi_2$, with $\sin\phi \approx \phi$.

2. $n_1\left(\dfrac{h}{x} + \dfrac{h}{r}\right) = n_2\left(\dfrac{h}{r} - \dfrac{h}{y}\right)$

   Substitute step B.

3. $\dfrac{n_1}{x} + \dfrac{n_2}{y} = \dfrac{n_2 - n_1}{r}$

   Divide by h and collect the r terms on the right (Ghatak 4.4). h **has gone**: every paraxial ray from O, at any height, reaches the same I. So I is an image.

Step D: put in the signs

1. $u = -x$ (object on the left), $\quad v = +y, \quad R = +r$ (C on the right)

   Sign rule (P4).

2. $-\dfrac{n_1}{u} + \dfrac{n_2}{v} = \dfrac{n_2 - n_1}{R}$

   Replace x by $-u$, y by v, r by R.

RESULT · SINGLE SPHERICAL SURFACE (GHATAK 4.5)

$$\frac{n_2}{v} - \frac{n_1}{u} = \frac{n_2 - n_1}{R}$$

n_1 = medium the light comes **from**, n_2 = medium it goes **into**. This formula holds for every case (convex or concave surface, real or virtual object or image) if you put in the signs.

SANITY CHECKS

• Flat surface ($R \to \infty$): $\frac{n_2}{v} = \frac{n_1}{u}$, so $v = \frac{n_2}{n_1}u$. A coin 12 cm deep in water ($n_1 = 4/3$, $n_2 = 1$) appears at $v = \frac{1}{4/3}(-12) = -9$ cm, i.e. 9 cm deep. This is the **apparent depth** from Plus Two.

• $n_1 = n_2$: the right side is 0 and $v = u$. With no change of medium, nothing happens.

Worked example: three sign cases

Glass ($n_2 = 1.5$), light coming from air ($n_1 = 1$).

1. **(a)** Convex towards the light, $R = +10$, real object $u = -30$: $\frac{1.5}{v} = \frac{0.5}{10} + \frac{1}{-30} = 0.05 - 0.0333 = 0.0167$, so $v = +90$ cm
   $v > 0$: real image inside the glass, 90 cm from P.

2 **(b)** Same surface, but a converging beam heading for a point 20 cm to the right of P, so a **virtual object** with $u = +20$: $\frac{1.5}{v} = 0.05 - \frac{1}{20} = 0$, so $v = \infty$

The refracted rays become parallel.

3 **(c)** Concave towards the light, $R = -10$, $u = -30$: $\frac{1.5}{v} = \frac{0.5}{-10} - \frac{1}{30} = -0.0833$, so $v = -18$ cm

$v < 0$: a virtual image on the left, 18 cm from P.

EXAM TRAPS

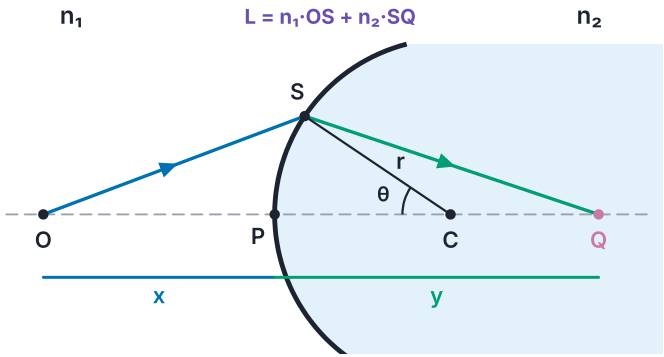
- Swapping n_1 and n_2. n_1 is always the side the light comes from.
- Writing $\frac{n_1}{u} + \frac{n_2}{v}$ with Cartesian signs. That is a different convention (P4).
- Forgetting the minus in $-\frac{n_1}{u}$ when u itself is negative. Write it out: $-\frac{1}{(-30)} = +\frac{1}{30}$.

In the model paper: **Q25** (PYQ N19 Q18), **Q32**, **Q34**.

T8 The same formula from Fermat (optical path route)

Ghatak Ex. 3.3. A 10-mark essay favourite: it links

Unit 1 to Unit 2.



Fermat route (Ghatak Fig. 3.10). Move S around the surface by changing θ. The ray is real when L is stationary: dL/dθ = 0.

Same surface: centre C, radius r, object O at distance x from P. Pick any point S on the surface, where CS makes angle θ with the axis. Let Q be any point on the axis, distance y from P, in medium n_2. Fermat says: the ray O → S → Q is real if $L = n_1\,OS + n_2\,SQ$ is stationary as S moves, i.e. $dL/d\theta = 0$.

Step A: length OS

1. $OS^2 = (x+r)^2 + r^2 - 2(x+r)\,r\cos\theta$
   Cosine rule in triangle OCS: $OC = x + r$, $CS = r$, angle at C is θ.

2. $\cos\theta \approx 1 - \dfrac{\theta^2}{2}$ gives $OS^2 \approx x^2 + r(x+r)\theta^2$
   Small angle (M1). Expand: $(x+r)^2 + r^2 - 2(x+r)r = x^2$, and the θ^2 part is $(x+r)r\theta^2$.

3. $OS \approx x + \dfrac{r(x+r)}{2x}\theta^2 = x + \tfrac{1}{2}r^2\left(\dfrac{1}{r} + \dfrac{1}{x}\right)\theta^2$
   Binomial shortcut (M4): $\sqrt{x^2 + k\theta^2} \approx x + \dfrac{k\theta^2}{2x}$.

Step B: length SQ

1. $SQ \approx y - \frac{1}{2}r^2\left(\frac{1}{r} - \frac{1}{y}\right)\theta^2$

   Same steps in triangle SCQ, where $CQ = y - r$ and the angle at C is $180^\circ - \theta$, so the cosine changes sign.

Step C: total optical path

1. $L = n_1 OS + n_2 SQ \approx (n_1 x + n_2 y) + \frac{1}{2}r^2\left[\frac{n_1}{x} + \frac{n_2}{y} - \frac{n_2 - n_1}{r}\right]\theta^2$

   Add the two, collect the θ^2 terms (Ghatak 3.7).

Step D: apply Fermat

1. $\frac{dL}{d\theta} = r^2\left[\frac{n_1}{x} + \frac{n_2}{y} - \frac{n_2 - n_1}{r}\right]\theta = 0$

   Stationary path (Ghatak 3.8).

2. Either $\theta = 0$ (the ray along the axis, which always works), or the bracket is zero

   For **all** rays from O to reach Q, it must hold for every θ, so the bracket must be 0.

3. $\frac{n_1}{x} + \frac{n_2}{y_0} = \frac{n_2 - n_1}{r}$

   The special distance $y = y_0$ where this happens is the image I (Ghatak 3.9). Signs $u = -x$, $v = y_0$, $R = r$ give $\frac{n_2}{v} - \frac{n_1}{u} = \frac{n_2 - n_1}{R}$ again.

THE IDEA

At the image, L does not depend on θ (to order θ^2). Every paraxial ray takes the **same time** to get from O to I. That is the "perfect imaging" condition of T5.

Minimum or maximum? (Ghatak, after eq. 3.10)

1. $\frac{d^2L}{d\theta^2} = r^2\left[\frac{n_1}{x} + \frac{n_2}{y} - \frac{n_2 - n_1}{r}\right] = r^2 n_2\left(\frac{1}{y} - \frac{1}{y_0}\right)$
   Replace $\frac{n_2 - n_1}{r} - \frac{n_1}{x}$ by $\frac{n_2}{y_0}$ using step D.

2. Q beyond the image ($y > y_0$): negative, so a **maximum**. Q before it ($y < y_0$): positive, so a **minimum**. At $y = y_0$: constant (stationary)
   Second-derivative test (M3). A real example where the actual ray takes a **maximum** time.

Virtual images by Fermat (Ghatak Ex. 3.4)

If the refracted rays spread out, they only *appear* to come from a point I on the object's side. The condition is that $n_1\,OS - n_2\,SI$ is the same for all S.

1. Take any far point B on a refracted ray. $L = n_1 OS + n_2 SB$
   The real optical path from O to B.

2. $L = (n_1 OS - n_2 SI) + n_2(IS + SB)$
   Add and subtract $n_2\,SI$.

3. If the first bracket is the same for all S, Fermat needs only $IS + SB$ stationary
   A sum of two lengths is least when I, S, B are on one straight line.

4. So each refracted ray lies on a straight line through I: it **appears** to come from I
   This is what a virtual image means. A minus sign in the optical path means "counted backwards".

> **CHECK YOURSELF**
>
> **In the Fermat route, why must the bracket be zero, and not just $\theta = 0$?**
>
> $\theta = 0$ is only the ray along the axis. For an image, rays at **all** small angles must reach the same point, so $dL/d\theta = 0$ must hold for every θ. That forces the bracket to vanish.

In the model paper: Q39, Q31.

T9 Foci of one surface, and surfaces in a row

> Two small ideas we need before lenses.

Focal lengths of a single surface

1. First focus F_1: the object point whose image is at infinity. Put $v = \infty$, $u = f_1$: $-\frac{n_1}{f_1} = \frac{n_2 - n_1}{R}$, so $f_1 = -\frac{n_1 R}{n_2 - n_1}$
   Rays from F_1 come out parallel.

2. Second focus F_2: the image of an object at infinity. Put $u = -\infty$, $v = f_2$: $\frac{n_2}{f_2} = \frac{n_2 - n_1}{R}$, so $f_2 = \frac{n_2 R}{n_2 - n_1}$
   Parallel rays meet at F_2.

3. $\dfrac{f_2}{f_1} = -\dfrac{n_2}{n_1}$

   Divide. For one surface $|f_1| \neq |f_2|$: the two focal lengths are **not** equal, unlike a lens in air.

Example: $n_1 = 1$, $n_2 = 1.5$, $R = +10$ cm: $f_1 = -20$ cm, $f_2 = +30$ cm.

Surfaces in a row: the image becomes the next object

THE IDEA (GHATAK EX. 4.1)

When light meets several surfaces, solve them **one at a time**. The image made by surface 1 is the object for surface 2. Measure the new u from surface 2's own pole.

RESULT · VIRTUAL OBJECT

If rays heading towards an image are caught by the next surface **before** they meet, that image lies to the right of the next surface. Then it acts as a **virtual object** with $u > 0$.

1. Glass rod, $n = 1.5$, first surface $R_1 = +15$ cm, object at $u = -40$ cm: $\dfrac{1.5}{v} = \dfrac{0.5}{15} - \dfrac{1}{40} = 0.00833$, so $v = +180$ cm

   Surface 1 alone would form an image 180 cm inside the glass.

2. Second surface 30 cm further on, $R_2 = -25$ cm. The image is $180 - 30 = 150$ cm to its right, so $u = +150$ (virtual object)

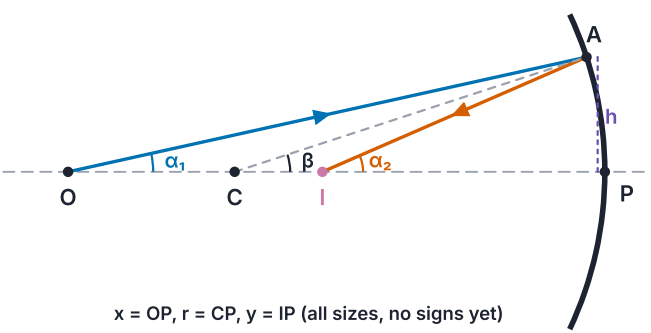
   The rays meet surface 2 before they can converge.

3. Now glass → air: $\frac{1}{v} - \frac{1.5}{150} = \frac{1 - 1.5}{-25} = 0.02$, so $v = +33.3$ cm
   Final real image 33.3 cm beyond the second surface.

In the model paper: Q13 , Q12 . This "image becomes object" idea is exactly how the lens formula is built in T11.

T10 Reflection at a spherical mirror

> Same method, with the law of reflection instead of Snell. Two routes: geometry, and Fermat.



Concave mirror (Ghatak Fig. 4.3 style). Grey dashed CA is the normal at A. The incident ray OA and reflected ray AI make equal angles with it.

Geometry route (Ghatak §4.3)

1. $\phi_1 = \beta - \alpha_1$
   Triangle OAC. β is exterior at C, so $\beta = \alpha_1 + \phi_1$.

2. $\phi_2 = \alpha_2 - \beta$
   Triangle CAI. α_2 is exterior at I, so $\alpha_2 = \beta + \phi_2$.

3. $\phi_1 = \phi_2$ gives $2\beta = \alpha_1 + \alpha_2$
   Law of reflection, $i = r$.

4. $\dfrac{2}{r} = \dfrac{1}{x} + \dfrac{1}{y}$

   Small angles: $\beta \approx \dfrac{h}{r}$, $\alpha_1 \approx \dfrac{h}{x}$, $\alpha_2 \approx \dfrac{h}{y}$, then cancel h (Ghatak 4.6).

5. Signs: O, C and I are all on the left, so $u = -x$, $v = -y$, $R = -r$: $\dfrac{1}{u} + \dfrac{1}{v} = \dfrac{2}{R}$

   Ghatak 4.7. With $f = R/2$ this is the Plus Two formula $\dfrac{1}{v} + \dfrac{1}{u} = \dfrac{1}{f}$.

RESULT · MIRROR EQUATION

$$\frac{1}{u} + \frac{1}{v} = \frac{2}{R} = \frac{1}{f}$$

Concave mirror: $R < 0$, $f < 0$. Convex mirror: $R > 0$, $f > 0$.

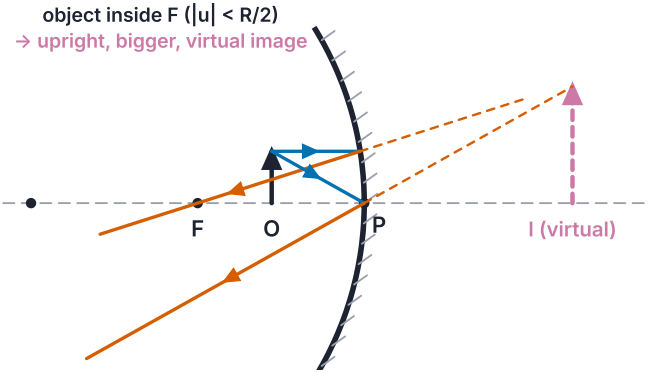
SHORTCUT: A MIRROR IS "REFRACTION WITH $n_2 = -n_1$" (GHATAK §4.3)

Put $n_2 = -n_1$ in the surface formula:

$$\frac{-n_1}{v} - \frac{n_1}{u} = \frac{-2n_1}{R}$$

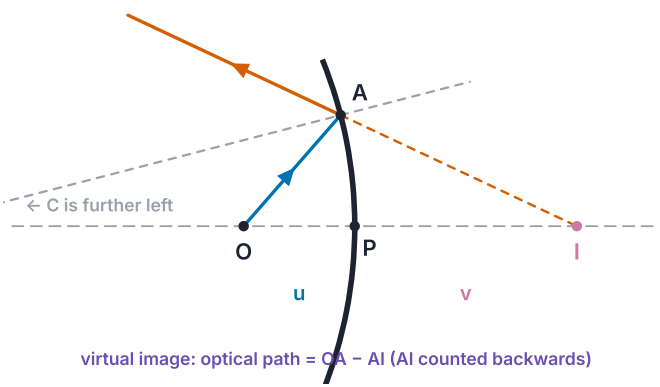
Divide by $-n_1$: $\dfrac{1}{v} + \dfrac{1}{u} = \dfrac{2}{R}$. The mirror equation.

Why it works: Snell's law $n_1 \sin\phi_1 = -n_1 \sin\phi_2$ gives $\phi_2 = -\phi_1$, i.e. the light turns back with equal angle: reflection.



Concave mirror with the object closer than F. The reflected rays (orange) spread out; extended backwards (dashed) they meet behind the mirror at I. Numbers used: $u = -50$, $f = -90$ → $v = +112.5$, $m = +2.25$.

Fermat route: object closer than $R/2$ (PYQ N20 Q25)



Fermat for the concave mirror with object inside $R/2$ (PYQ N20 Q25). The image is virtual, so the "path" to I is measured backwards along the dashed line.

Concave mirror, centre C at distance r, object O at distance $x < r/2$ from P. The image Q is virtual, behind the mirror, at distance y. S is a point on the mirror where CS makes angle θ with the axis.

1. $OS \approx x + \frac{1}{2}r^2\left(\frac{1}{x} - \frac{1}{r}\right)\theta^2$
   Cosine rule in triangle OCS ($CO = r - x$), then the small-angle and binomial steps of T8.

2. $SQ \approx y + \frac{1}{2}r^2\left(\frac{1}{y} + \frac{1}{r}\right)\theta^2$
   Triangle QCS with $CQ = r + y$.

3. Virtual image, so use $L = OS - SQ \approx (x - y) + \frac{1}{2}r^2\left[\frac{1}{x} - \frac{1}{y} - \frac{2}{r}\right]\theta^2$
   As in T8 (virtual images): the reflected ray only appears to come from Q, so SQ is counted backwards.

4. $\frac{dL}{d\theta} = 0$ for all θ: $\frac{1}{x} - \frac{1}{y} = \frac{2}{r}$
   The bracket must vanish.

5. Signs $u = -x$, $v = +y$ (behind), $R = -r$: $\frac{1}{u} + \frac{1}{v} = \frac{2}{R}$
   Check: $-\frac{1}{x} + \frac{1}{y} = -\frac{2}{r}$. ✓ The same mirror equation.

Note: $\frac{1}{y} = \frac{1}{x} - \frac{2}{r} > 0$ only when $x < \frac{r}{2}$. That is exactly why the image is behind the mirror (virtual) when the object is inside the focus.

EXAM TRAPS

- Mirror: $\frac{1}{v}+\frac{1}{u}$. Lens: $\frac{1}{v}-\frac{1}{u}$. Do not mix them.
- A concave mirror has $R<0$ and $f<0$ in this convention.
- Mirror magnification is $m=-v/u$ (T15), lens magnification is $m=+v/u$.

CHECK YOURSELF

1. Concave mirror, $R=-60$ cm, object at $u=-80$ cm. Image?

$\frac{1}{v}=\frac{2}{R}-\frac{1}{u}=-\frac{1}{30}+\frac{1}{80}=\frac{-8+3}{240}$, so $v=-48$ cm: real, in front of the mirror.

2. Same mirror, object at $u=-20$ cm?

$\frac{1}{v}=-\frac{1}{30}+\frac{1}{20}=\frac{1}{60}$, so $v=+60$ cm: virtual, behind the mirror.

In the model paper: **Q33**, **Q14**, **Q26** (PYQ N20 Q25), **Q35**.

UNIT 3 · GHATAK §4.4–4.5, 4.7

The thin lens

Two curved surfaces, very close together. Apply T7 twice.

T11 Thin lens formula and lens-maker's formula

> Where $\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$ and $\frac{1}{f} = (n-1)(\frac{1}{R_1} - \frac{1}{R_2})$ come from.

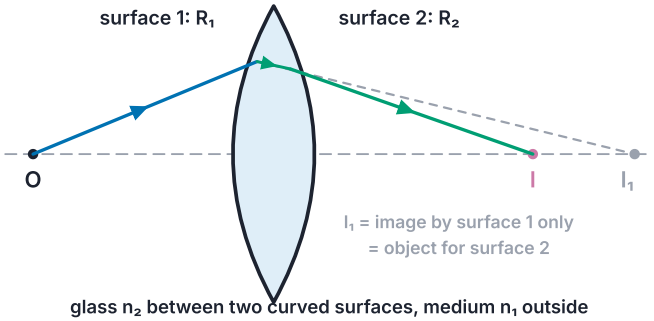
RESULT · DEFINITION OF A THIN LENS (PYQ N24 Q1)

A lens is a piece of transparent material bounded by two spherical surfaces. It is **thin** when its thickness is negligible compared with the object distance, the image distance and the radii of curvature. Then all distances can be measured from one point, the centre of the lens.

Signs of R_1, R_2 (light from the left)

Biconvex	$R_1 > 0$, $R_2 < 0$
Biconcave	$R_1 < 0$, $R_2 > 0$
Meniscus	both the same sign (both + or both −)
Plano-convex	flat side has $R = \infty$, so $1/R = 0$

Rule: R is + if that surface's centre is on the right (P4).



glass n_2 between two curved surfaces, medium n_1 outside

A thin lens = two curved surfaces very close together. Surface 1 alone would form an image I_1. Surface 2 takes I_1 as its object and forms the final image I.

Lens of index n_2 in a medium of index n_1. Object O at distance u.

1. Surface 1 alone (from n_1 into n_2): $\frac{n_2}{v'} - \frac{n_1}{u} = \frac{n_2 - n_1}{R_1}$
   T7 formula. v' is where surface 1 alone would form the image I_1 (Ghatak 4.8).

2. Surface 2 (from n_2 back into n_1), object I_1 at v': $\frac{n_1}{v} - \frac{n_2}{v'} = \frac{n_1 - n_2}{R_2}$
   Same T7 formula, with three swaps: its object distance "u" is v'; the light now comes *from* n_2 and goes *into* n_1; its radius is R_2. The lens is thin, so v' is measured from the same point (Ghatak 4.9).

3. Add the two equations: $\frac{n_1}{v} - \frac{n_1}{u} = (n_2 - n_1)\left(\frac{1}{R_1} - \frac{1}{R_2}\right)$
   The $\frac{n_2}{v'}$ terms **cancel**, so the middle image disappears.

4. $\frac{1}{v} - \frac{1}{u} = (n-1)\left(\frac{1}{R_1} - \frac{1}{R_2}\right)$, where $n = \frac{n_2}{n_1}$
   Divide by n_1. n is the index of the lens **relative to** its surroundings (Ghatak 4.10).

RESULT · THIN LENS (GHATAK 4.11, 4.12)

$$\frac{1}{v} - \frac{1}{u} = \frac{1}{f}, \qquad \frac{1}{f} = (n-1)\left(\frac{1}{R_1} - \frac{1}{R_2}\right)$$

The second formula is the **lens-maker's formula**. For a lens in air, n is just the glass index.

Worked example (PYQ N25 Q13 type)

Biconvex lens, $n = 1.5$, radii 50 cm and 100 cm, in air.

1. $R_1 = +50$, $R_2 = -100$
   Biconvex signs.
2. $\frac{1}{f} = 0.5\left(\frac{1}{50} + \frac{1}{100}\right) = 0.5 \times 0.03 = 0.015$, so $f = 66.7$ cm
   Note $-\frac{1}{R_2} = -\frac{1}{-100} = +\frac{1}{100}$.
3. Real object, image wanted 60 cm away: since $60 < f$, the image must be on the **same side** as the object (virtual), so $v = -60$
   A converging lens with a real object gives a real image only beyond f (P3 table).

4. $\frac{1}{u} = \frac{1}{v} - \frac{1}{f} = -\frac{1}{60} - 0.015 = -0.03167$, so $u = -31.6$ cm

   Object 31.6 cm in front of the lens: inside F_1, magnifier case. ✓

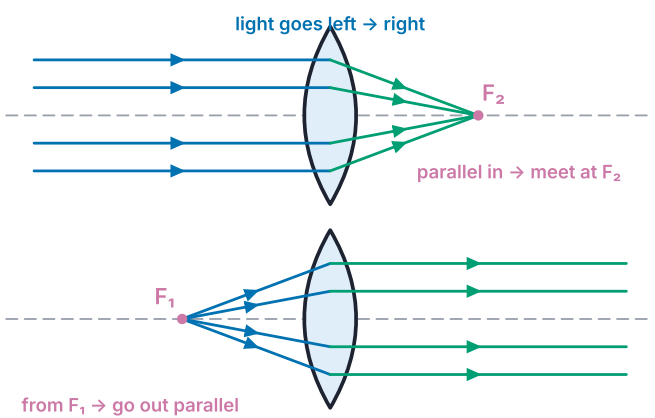
EXAM TRAP

Losing the sign in $\left(\frac{1}{R_1} - \frac{1}{R_2}\right)$. For a biconvex lens the two terms **add**.

In the model paper: **Q15** (PYQ N24 Q1), **Q16**, **Q40**, **Q27** (PYQ N25 Q13).

T12 Principal foci and focal length

> The two special points of a lens, and what changes when the lens is put in water.



Top: **second principal focus F_2**: rays parallel to the axis meet there after the lens. Bottom: **first principal focus F_1**: rays starting there come out parallel. For a thin lens in air both are a distance f from the lens.

RESULT · DEFINITIONS (PYQ N24 Q2)

First principal focus F_1: the point on the axis such that rays starting from it (or heading towards it) come out of the lens **parallel to the axis**. Its image is at infinity.

Second principal focus F_2: the point where rays coming in **parallel to the axis** meet after the lens (or seem to come from, for a diverging lens).

Focal lengths: f_1 = distance of F_1 from the lens, f_2 = distance of F_2, both with signs.

Lens with different media on the two sides (Ghatak §4.5)

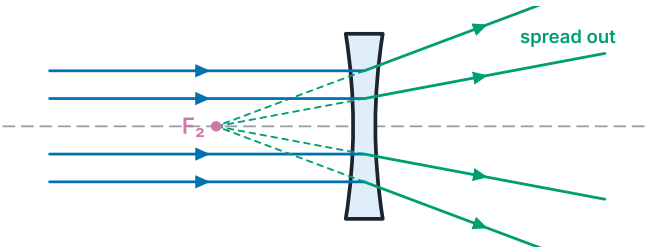
Medium n_1 on the left, lens n_2, medium n_3 on the right. Adding the two surface equations as in T11 gives:

1. $$\frac{n_3}{v} - \frac{n_1}{u} = \frac{n_2 - n_1}{R_1} + \frac{n_3 - n_2}{R_2}$$
   Ghatak 4.14. Call the right-hand side D.

2. F_1: put $v = \infty$, $u = f_1$: $f_1 = -\dfrac{n_1}{D}$
   Ghatak 4.15.

3. F_2: put $u = -\infty$, $v = f_2$: $f_2 = \dfrac{n_3}{D}$
   Ghatak 4.16.

4. Same medium both sides ($n_3 = n_1$): $f_2 = -f_1 = f$
   Ghatak 4.13. So for a lens in air F_1 and F_2 are the same distance f from the lens, on opposite sides.

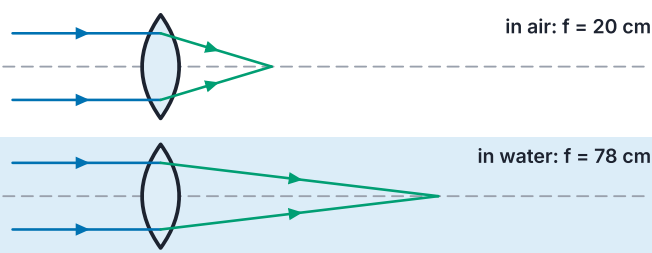


Concave (diverging) lens: parallel rays spread out as if coming from F_2, which is on the incoming side. So f is negative.

CONVERGING OR DIVERGING?

$f > 0$ (so $f_2 > 0$, $f_1 < 0$): **converging**. $f < 0$: **diverging**. In air, a biconvex glass lens has $n - 1 > 0$ and $\frac{1}{R_1} - \frac{1}{R_2} > 0$, so it converges.

Glass lens in water (PYQ N25 Q1)



Same glass lens ($n = 1.5$, $R_1 = 20$, $R_2 = -20$ cm). In water ($n = 1.33$) it still converges, but much more weakly, because glass is only a little "slower" than water.

1. Relative index $n = \frac{n_{glass}}{n_{water}} = \frac{1.5}{1.33} = 1.128$
   In T11, $n = n_2/n_1$: lens relative to its surroundings.

2. $n - 1 = 0.128 > 0$, and $\left(\frac{1}{R_1} - \frac{1}{R_2}\right) > 0$ for biconvex, so $f > 0$
   It is **still converging**.

3 $\frac{f_{water}}{f_{air}} = \frac{1.5-1}{1.128-1} = \frac{0.5}{0.128} \approx 3.9$

The curvature factor cancels. The focal length becomes about 4 times longer, so the lens is much weaker.

EXAM TRAP

If the lens is **less dense** than the surroundings ($n_{lens} < n_{medium}$, e.g. an air bubble in water, or glass of 1.5 in a liquid of 1.6), then $n < 1$: a biconvex lens **diverges** and a biconcave one converges.

REMIND ME · POWER (PLUS TWO)

$P = \frac{1}{f}$ with f in **metres**; unit dioptre (D).

Converging: $P > 0$. Diverging: $P < 0$.

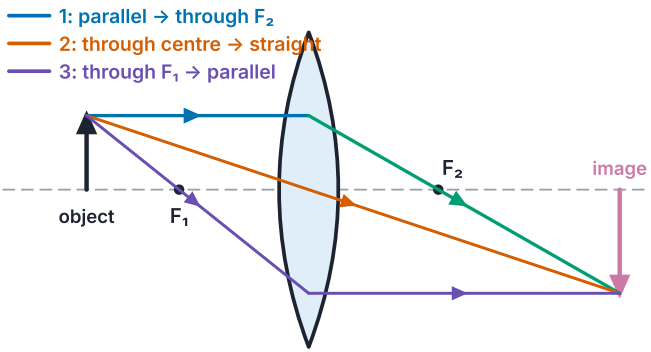
Example: $f = -25$ cm gives $P = 1/(-0.25) = -4$ D. (Not in Ghatak's sections, but useful.)

In the model paper: **Q17** (PYQ N24 Q2), **Q18** (PYQ N25 Q1), **Q41** , **Q24** .

T13 Finding the image by drawing

Three easy rays. Use them to sketch every answer

and to check your signs.



Three easy rays from the top of the object. Where they meet is the top of the image. (Numbers: $f = 70$, $u = -120 \to v \approx 168$.)

RESULT · THREE RULES (GHATAK §4.7)

1. A ray parallel to the axis goes through F_2 (or seems to come from F_2 for a diverging lens).
2. A ray through the centre of the lens goes straight on, undeviated.
3. A ray through F_1 (or heading to F_1) comes out parallel to the axis.

Any two of them locate the top of the image.

WHY THE MIDDLE RAY IS STRAIGHT

Near its centre a thin lens is like a thin parallel slab of glass. A slab only shifts a ray sideways, and for a thin lens that shift is negligible.

CHECK YOURSELF

Object inside F_1 of a convex lens: where do rays 1 and 2 meet?

They spread out after the lens and do not meet. Extended backwards they meet on the object's side: a virtual, upright, bigger image.

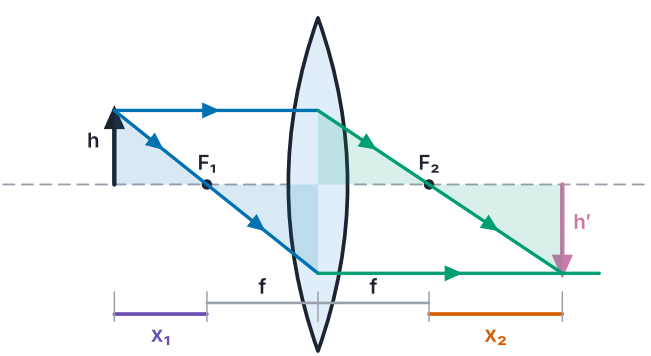
In the model paper: Q19 , Q38 .

UNIT 4 · GHATAK §4.6–4.7

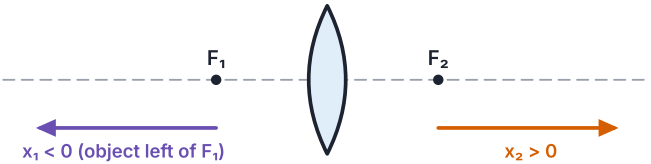
Newton's formula and magnification

T14 Newton's formula

> The lens formula again, with distances measured from the foci instead of the lens.



Newton's formula. x_1 is measured from F_1 to the object, x_2 from F_2 to the image (not from the lens!). Blue triangles are similar; green triangles are similar.



x_1 is a coordinate measured **from** F_1, x_2 is measured **from** F_2, with the usual signs.

Object of height h is a distance $|x_1|$ to the left of F_1. Image of height h' (below the axis) is $|x_2|$ to the right of F_2. The lens is in air, so both focal distances are f.

Step A: two pairs of similar triangles (sizes only)

1. Blue pair, meeting at F_1: $\dfrac{|h|}{|x_1|} = \dfrac{|h'|}{f}$

   The ray through F_1. One triangle has the object as one side, the other has the lens (height $|h'|$, because the ray then goes parallel at the image's height).

2. Green pair, meeting at F_2: $\dfrac{|h|}{f} = \dfrac{|h'|}{|x_2|}$

   The parallel ray: it leaves the lens at height $|h|$ and passes through F_2 to the image top.

3. Both give $\dfrac{|h'|}{|h|}$: $\dfrac{f}{|x_1|} = \dfrac{|x_2|}{f}$ so $|x_1|\,|x_2| = f^2$

   Set the two expressions for the size ratio equal.

Step B: signs

1. Object left of F_1: $x_1 < 0$. Image right of F_2: $x_2 > 0$

   Coordinates measured from each focus, with the usual sign rule.

2. $x_1\,x_2 = -f^2$ (general form: $x_1 x_2 = f_1 f_2$)
   The product of a negative and a positive number. With $f_1 = -f$, $f_2 = f$: $f_1 f_2 = -f^2$ (Ghatak 4.19, 4.20).

RESULT · NEWTON'S FORMULA

$$x_1\,x_2 = f_1 f_2 = -f^2$$

(lens in air, Cartesian signs)

In sizes only (as in many guides, and your old notes): $|x_1|\,|x_2| = f^2$. If a question writes $x_1 x_2 = f^2$, it means sizes. Say so in one line, then continue.

WHAT IT TELLS YOU

x_1 and x_2 have opposite signs. If the object is to the left of F_1, the image is to the right of F_2. Move the object closer to F_1 ($|x_1|$ smaller) and the image runs away ($|x_2|$ larger).

1. Check with numbers: $f = 75$ cm, object at $u = -100$ cm. Lens formula: $\frac{1}{v} = \frac{1}{75} - \frac{1}{100}$, so $v = +300$ cm
   Ghatak problem 4.1.

2. $x_1 = u - f_1 = -100 - (-75) = -25, \quad x_2 = v - f_2 = 300 - 75 = +225$
   Distance from the focus = (distance from the lens) − (focus position).

3. $x_1 x_2 = -25 \times 225 = -5625 = -(75)^2$ ✓
   Newton agrees with the lens formula.

EXAM TRAP

x_1 and x_2 are measured from the **foci**, not from the lens.

In the model paper: Q36 , Q21 .

T15 Lateral magnification

How big is the image, and is it upside down?

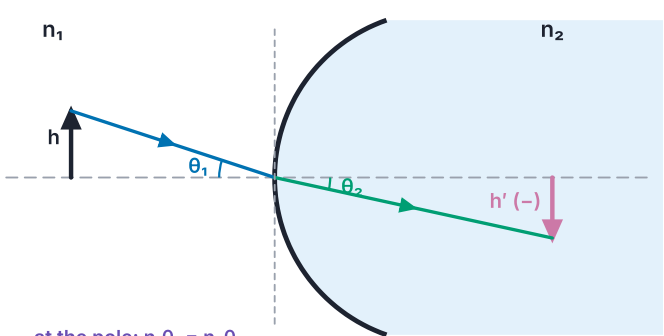
RESULT · DEFINITION (PYQ N24 Q3)

Lateral (transverse) magnification is the ratio of the image height to the object height, both measured perpendicular to the axis with signs:

$$m = \frac{h'}{h}$$

$m > 0$: upright (erect). $m < 0$: inverted. $|m| > 1$: bigger. $|m| < 1$: smaller.

For one refracting surface (Ghatak 4.23)



Magnification at a single surface: follow the ray that hits the pole. $\theta_1 \approx h/(-u)$, $\theta_2 \approx -h'/v$, and Snell (small angle) gives $m = h'/h = n_1 v/(n_2 u)$.

1. Ray from the object top to the pole P. Angles: $\theta_1 \approx \dfrac{h}{-u}, \quad \theta_2 \approx \dfrac{-h'}{v}$

   Small angles (M1). $-u$ and $-h'$ are positive sizes, because $u < 0$ and the image is inverted.

2. $n_1\theta_1 = n_2\theta_2$: $\quad n_1\dfrac{h}{-u} = n_2\dfrac{-h'}{v}$

   Snell at P (where the normal is the axis itself), small angles.

3. $m = \dfrac{h'}{h} = \dfrac{n_1 v}{n_2 u}$

   Rearrange. The minus signs cancel.

For a thin lens: multiply the two surfaces (Ghatak 4.24)

1. $m_1 = \dfrac{n_1 v'}{n_2 u}, \quad m_2 = \dfrac{n_2 v}{n_1 v'}$

   Surface formula, used twice. For surface 2 the media are swapped and its object is at v'.

2. $m = m_1 m_2 = \dfrac{v}{u}$

   The final image is m_2 times the middle image, which is m_1 times the object. v', n_1, n_2 all cancel.

3. Using Newton: $m = -\dfrac{f_1}{x_1} = -\dfrac{x_2}{f_2}$, i.e. $m = \dfrac{f}{x_1} = -\dfrac{x_2}{f}$ in air

   From the similar triangles of T14, with signs (Ghatak 4.21).

> **MIRROR**
>
> Put $n_2 = -n_1$ in $m = \dfrac{n_1 v}{n_2 u}$: $m = -\dfrac{v}{u}$. That is the Plus Two mirror formula.

RESULT · ALL MAGNIFICATION FORMULAS

Surface	$m = \dfrac{n_1 v}{n_2 u}$
Thin lens	$m = \dfrac{v}{u} = \dfrac{f}{x_1} = -\dfrac{x_2}{f}$
Mirror	$m = -\dfrac{v}{u}$

BONUS: LONGITUDINAL MAGNIFICATION

How much the image moves along the axis when the object moves a little. Differentiate $\dfrac{1}{v} - \dfrac{1}{u} = \dfrac{1}{f}$: $-\dfrac{dv}{v^2} + \dfrac{du}{u^2} = 0$, so $\dfrac{dv}{du} = \dfrac{v^2}{u^2} = m^2$. It is always positive: the image moves the same way as the object.

CHECK YOURSELF

1. $f = 75$ cm, $u = -100$, $v = +300$. Find m in two ways.

$m = v/u = -3$. Also $m = f/x_1 = 75/(-25) = -3$. ✓ Inverted, 3 times bigger.

2. $f = +20$ cm, object at $u = -12$ cm. Image and m?

$\dfrac{1}{v} = \dfrac{1}{20} - \dfrac{1}{12} = -\dfrac{1}{30}$, so $v = -30$ cm, and $m = \dfrac{-30}{-12} = +2.5$: virtual, upright, 2.5 times bigger (a magnifier).

In the model paper: **Q20** (PYQ N24 Q3), **Q42**, **Q22**, **Q23**, **Q37**.

One-page summary

Cover the right column and test yourself.

Optical path	$L = \int n\,ds = c\times$ time
Fermat	$\delta\int n\,ds = 0$: stationary (min, max or constant)
3 results	straight lines · $i = r$ · $n_1 \sin\theta_1 = n_2 \sin\theta_2$
Signs	from the pole; along the light +; up +; $R > 0$ if C is on the right
Paraxial	$\sin\phi \approx \tan\phi \approx \phi$, rays near the axis
Surface	$\frac{n_2}{v} - \frac{n_1}{u} = \frac{n_2 - n_1}{R}$
Surface foci	$f_1 = -\frac{n_1 R}{n_2 - n_1}$, $f_2 = \frac{n_2 R}{n_2 - n_1}$
Mirror	$\frac{1}{u} + \frac{1}{v} = \frac{2}{R} = \frac{1}{f}$ $(n_2 = -n_1)$
Thin lens	$\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$
Lens maker	$\frac{1}{f} = (n-1)\left(\frac{1}{R_1} - \frac{1}{R_2}\right)$, $n = \frac{n_{lens}}{n_{medium}}$
Mixed media	$\frac{n_3}{v} - \frac{n_1}{u} = \frac{n_2 - n_1}{R_1} + \frac{n_3 - n_2}{R_2}$
Foci (lens)	$f_1 = -\frac{n_1}{D}$, $f_2 = \frac{n_3}{D}$; in air $f_2 = -f_1 = f$
Newton	$x_1 x_2 = -f^2$; sizes: $\lvert x_1\rvert\lvert x_2\rvert = f^2$
m (surface)	$\frac{n_1 v}{n_2 u}$
m (lens)	$\frac{v}{u} = \frac{f}{x_1} = -\frac{x_2}{f}$
m (mirror)	$-\frac{v}{u}$

Exam question → topics to know

Every past-paper question on Module 1 (2019–2025), broken down into its simplest topics.

PYQ	QUESTION (SHORT)	KNOW
N24 Q1	Define a thin lens	T11

N24 Q2	Define first principal focus	T12 T13
N24 Q3	Define lateral magnification	T15 M2
N25 Q1	Glass lens in water: converging?	T12 T11 P1
N25 Q13	Biconvex 50/100 cm, image at 60 cm: find object	T11 P4 M6 P3
N20 Q1	Optical path = ? × refractive index	T1
N20 Q11	Three results from Fermat	T2 T3 T4
N20 Q25	Mirror equation by Fermat, object < R/2	T10 T8 M1 M4
N19 Q11	Fermat + elliptical reflector	T2 T5 M5
N19 Q18	Image condition at refracting surface	T7 T6 M1 M2
N22 Q1	Paraxial rays pass ___ the axis	T6
N22 Q11	Fermat's principle of extremum path	T2

Left out on purpose (not in this module's syllabus): matrix-method questions (N19 Q1, Q23, Q25; N20 Q2, Q6, Q12, Q18; N22 Q18), thick lens, two-lens combinations (N22 Q25), zone plates (N19 Q29, N20 Q30), fibre and optical direction cosines (N22 Q12, Q17).